

KOHLER Facility Intelligence

Detailed Prompt Documentation & AI-Assisted Development Record

Track 2: Commercial Smart Facility & Sustainability Manager

A professional record of the problem framing, prompt evolution, AI workflows, implementation decisions, guardrails and final system behavior used to develop the functional prototype.

Document field	Value
Submission track	Track 2 — Commercial Smart Facility & Sustainability Manager
Prototype	KOHLER Facility Intelligence
Architecture	React + Vite frontend with Node telemetry/ticket simulation backend
Core loop	Telemetry → anomaly detection → incident → maintenance dispatch → resolution → sustainability impact
Primary AI role	Design support, architecture reasoning, UX iteration, documentation and evidence-grounded local Copilot
Data status	Synthetic/local simulation estimates; not real facility measurements
Version	Detailed documentation — corrected PDF build

Important documentation convention

Where an exact historical prompt transcript was not preserved in the final artifact set, the evolution below is explicitly described as a **reconstructed design record** rather than presented as a verbatim quotation. Final prompts are recorded as implementation-oriented prompt specifications and are traceable to the corresponding feature/module.

1. Executive Summary

The solution was designed around the case study's closed-loop operational requirement: high-frequency facility telemetry should not merely be visualized; it should drive detection, prioritization, dispatch, resolution and measurable sustainability insight.

Core product thesis: convert raw restroom and facility signals into an operator workflow that answers five questions: What is happening? How severe is it? Where is it? What action should the facility team take? What resource impact is being avoided?

Requirement	Implemented response
Water flow + continuous leaks	Live simulated flow, leak injection, excess-flow analysis and projected water-loss estimates
Flush/usage + occupancy	Usage-aware utilization and hygiene/traffic signals
Sensor diagnostics	Sensor fault scenario and attention-state workflow
Predictive dispatch	Automatic simulated maintenance ticket creation tied to critical/attention incidents
Human control	Start/stop/pause/reset simulation plus manual ticket creation, status changes, resolution and removal
Facility intelligence	Terminal/floor/zone context and Facility Twin fixture view
Sustainability	Virtual water footprint, leak-loss estimates and reduction actions
AI assistance	Evidence-grounded local Copilot with anti-hallucination constraints

2. Prompt & Product Evolution

The project evolved through successive refinement. Early iterations focused on making the dashboard visually credible; later iterations shifted the system toward an operationally meaningful closed loop.

Version	Design focus	Major evolution
v0.1	Problem framing	Defined commercial smart-facility dashboard, telemetry, maintenance and sustainability objectives.
v0.2	Interaction foundation	Established dashboard navigation, facility context, cards and initial simulation concepts.
v0.3	Facility visualization	Introduced Facility Twin/restroom fixture representation and contextual views.
v0.4	Operational workflow	Expanded live operations, alerts and maintenance concepts.
v0.5	Reference-aligned UI	Refined visual hierarchy, cards, controls, dark mode and KOHLER-style presentation.
v0.6	Backend integration	Connected frontend to Node telemetry/ticket simulation and shared state.
v0.7–v0.8	Full interactive UI	Expanded pages, controls, analytics and simulation interaction.
v0.9	Simulation reliability	Made Simulation the master control for telemetry and simulated ticket generation.
v1.0	Submission architecture	Consolidated functional prototype and case-study workflow.
v1.1	Modular final	Separated reusable data, API, stream hook, components, pages and backend logic.

The strongest conceptual change was moving from a **dashboard that shows events** to a **facility operating system that closes the loop around events**.

3. Prompt Catalog

The following catalog records the principal prompt specifications used to reason about the system. They are organized by function so a reviewer can reproduce or extend the work.

P-01 — Case-study framing

Act as a product architect for a commercial smart-facility and sustainability platform. Translate the KOHLER Track 2 case into a concrete prototype. Prioritize high-frequency water flow, flush counts, occupancy, continuous leaks and sensor diagnostics. The output must close the loop from telemetry to anomaly detection, maintenance dispatch, human resolution and sustainability impact. Keep the prototype feasible for a local demo and make all synthetic data clearly identifiable.

P-02 — Operational dashboard

Design an operator-first facility dashboard for a high-footfall commercial environment. Show the current facility state, water consumption, active incidents, maintenance workload, utilization and sustainability impact. Avoid decorative metrics that do not support an operational decision. Every critical metric should answer what happened, where, severity, and what action is available.

P-03 — Simulation master control

Design the simulation as the master state machine. Starting the simulation must start both restroom telemetry and simulated maintenance-ticket activity. Stopping must freeze telemetry and prevent new simulated tickets. Pause must suspend progression without resetting state. Reset must return the scenario to a known baseline. Manual ticket operations must remain possible regardless of automatic simulation state.

P-04 — Leak detection

Model continuous water leakage as an operational anomaly rather than a one-time alert. Detect sustained abnormal flow, identify the affected fixture/zone, assign severity, estimate water lost over time, create an incident and dispatch a maintenance ticket. Ensure the same anomaly does not generate an uncontrolled duplicate ticket on every telemetry cycle.

P-05 — Severity model

Define explicit facility severity states. Normal represents healthy operation; attention represents elevated demand, degraded sensor quality or hygiene/usage concern; critical represents a confirmed high-impact condition such as continuous leak or severe failure. Severity must be visible in the UI and must drive dispatch behavior rather than being cosmetic.

P-06 — Maintenance tickets

Create a maintenance workflow that supports both automated and human-created tickets. Tickets require an identifier, source, location, issue, priority, status and timestamps. Operators must be able to create, change status, resolve and remove tickets. A resolved status must be explicitly represented and can be produced by simulation or manually.

P-07 — Analytics

Build analytics from the same telemetry stream used by the operational dashboard. Analyze water consumption, leak/excess-flow behavior, utilization, incident health, water by fixture type, projected leak water loss, maintenance workload and virtual water footprint. Do not invent measurements that are not supported by the simulation state.

P-08 — Sustainability

Create a virtual water-footprint view using transparent synthetic assumptions. Break the footprint into direct facility water use and avoidable loss signals such as leaks. Explain reduction opportunities, estimated avoided loss and the operational action required. Label all estimates as simulated/proxy values.

P-09 — Facility Twin

Create an interactive facility-twin view of instrumented restroom zones and fixtures. Support terminal/floor/zone context and fixture filtering. Fixture states should reflect live simulation conditions so that an operator can move from an aggregate alert to a physical-looking location.

P-10 — AI Copilot

Act as an evidence-grounded facility operations copilot. Answer only from current application state and explicitly available evidence. Summarize active incidents, explain water anomalies, identify affected fixtures, describe maintenance workload and suggest next operational actions. Never fabricate sensor values, ticket IDs, locations or historical events. State when evidence is insufficient.

P-11 — Reusable architecture

Refactor the prototype into reusable, editable modules. Keep facility data separate from UI, API calls separate from pages, stream handling in a dedicated hook, shared UI in components, and simulation logic in the backend. Avoid one giant component. New scenarios, fixtures, charts and ticket fields should be addable without rewriting unrelated pages.

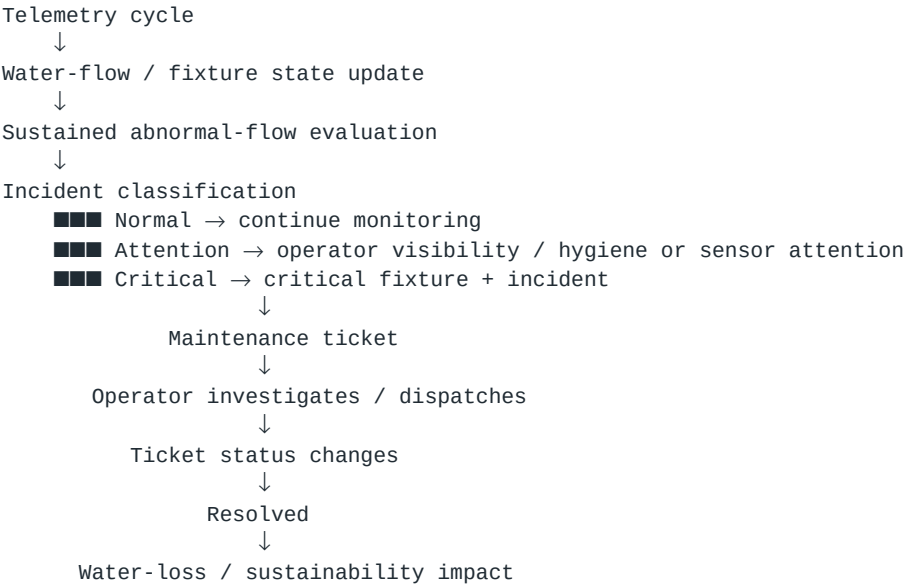
P-12 — AI/data guardrails

Treat all telemetry as synthetic simulation data. Never imply that simulated values are real-world measurements. Keep deterministic application state as the source of truth. Use AI for interpretation and assistance, not for inventing operational facts. If an answer cannot be grounded in application evidence, say so.

4. Detailed AI & System Workflows

The important design decision was to keep deterministic application logic responsible for operational state while using AI where interpretation, explanation and ideation add value.

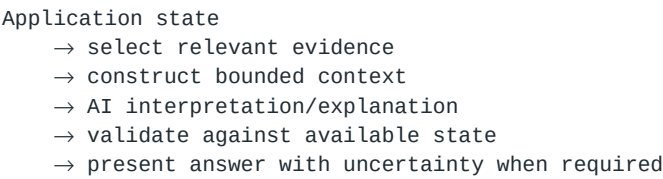
4.1 Closed-loop leak workflow



4.2 Ticket workflow

Action	Automatic simulation	Manual operator
Create	Scenario/incident can create ticket	Operator can create ticket
Update	Scenario may change operational state	Operator can change status/priority
Resolve	Scenario may mark an event resolved	Operator can explicitly resolve
Remove	Not used as a repeated telemetry action	Operator can remove ticket
Stop simulation	Stops new simulated tickets	Manual ticket controls remain available

4.3 AI evidence workflow



5. Prompt-to-Code Traceability

Prompt	Implementation area	Observable result
P-01	App + architecture	Track 2 requirements mapped to closed-loop product
P-02	pages/Overview.jsx	Facility summary and operational KPIs
P-03	pages/Simulation.jsx + server/index.mjs	Start/stop/pause/reset master simulation behavior
P-04	server/index.mjs + Analytics.jsx	Leak scenario, incident, ticket and projected loss
P-05	facility state / UI	Normal, attention and critical states
P-06	Maintenance.jsx + backend	Create, status change, resolve and remove
P-07	Analytics.jsx	Multi-dimensional water, leak, utilization and workload analysis
P-08	Sustainability.jsx	Virtual water footprint and reduction actions
P-09	FacilityTwin.jsx + data/facility.js	Instrumented restroom/fixture map and filters
P-10	AI Copilot area + application state	Evidence-grounded operational assistance
P-11	src/ modular structure	Reusable pages, components, hooks, API and data modules
P-12	README + runtime behavior	Synthetic-data disclosure and evidence boundaries

Final repository structure

```
src/
  App.jsx
  main.jsx
  data/facility.js
  services/api.js
  hooks/useFacilityStream.js
  components/Common.jsx
  pages/
    Overview.jsx
    LiveOperations.jsx
    FacilityTwin.jsx
    Analytics.jsx
    Maintenance.jsx
    Sustainability.jsx
    Simulation.jsx
    OtherPages.jsx
  styles.css

server/
  index.mjs
```

6. Analytics Intelligence Specification

Analytics was intentionally expanded beyond a single water-consumption graph. The goal is to make leaks and abnormal usage analytically visible, quantify their impact, and connect those measurements to operational decisions.

Analysis	Question answered	Example output
Water trend	Is demand changing?	Live flow trend / baseline comparison
Leak/excess-flow trend	Is abnormal flow persistent?	Leak intensity and duration
Projected leak loss	How much water could be wasted?	Estimated loss over time
Fixture utilization	Which fixture types receive the most demand?	Usage distribution
Incident health	How healthy is the facility?	Normal / attention / critical mix
Water by fixture type	Where is consumption concentrated?	Toilet / urinal / sink / other
Maintenance workload	Where is operational effort accumulating?	Open/critical/resolved workload
Virtual water footprint	What is the resource impact?	Simulated direct use + avoidable loss

6.1 Insight generation

A useful insight is not simply “water use increased.” The intended analysis combines magnitude, persistence, location, operational consequence and possible action. For example: sustained excess flow at a specific fixture increases projected loss; the fixture becomes critical; a maintenance ticket is dispatched; resolving the ticket stops the simulated loss; the sustainability view can then estimate avoided waste.

6.2 Virtual water footprint

The prototype treats the footprint as a **virtual operational estimate**, not a certified lifecycle assessment. Its purpose is to make water-loss consequences legible during a demonstration and to show how telemetry can be connected to sustainability decisions.

7. UX, Simulation Controls & Reliability Requirements

The development iterations identified several interaction requirements that are important for a live demo.

Control	Required behavior
Start simulation	Starts restroom telemetry and ticket simulation together.
Stop simulation	Stops both; freezes telemetry progression and prevents new simulated tickets.
Pause	Temporarily suspends progression without discarding current state.
Reset scenario	Returns simulation to a known baseline.
Scenario cards	Normal, High Traffic, Leak Injection and Sensor Fault alter the simulated state.
Manual tickets	Create, update, resolve and remove independently of automatic generation.
Resolved state	Explicit ticket status; may be generated by simulation or set manually.
Context dropdowns	Terminal, floor and zone are clickable controls and persist through the context API.
Dark mode	All major controls and pages remain readable and interactive in dark theme.

7.1 Failure modes explicitly considered

- Frontend loading failure caused by CSS/build issues must not be confused with application-state failure.
- Simulation must not continue generating events after Stop.
- Repeated telemetry cycles must not create unlimited duplicate maintenance tickets for one continuous incident.
- Analytics must tolerate a stopped simulation and retain the last known state.
- AI assistance must not invent unavailable telemetry.
- Synthetic data must be clearly distinguished from real facility measurements.

8. KOHLER Evaluation-Criteria Mapping

Criterion	How the solution addresses it
Approach & Innovation — 45%	Closed-loop operational intelligence; leak persistence analysis; automatic dispatch; Facility Twin; sustainability linkage; evidence-grounded AI assistance; modular architecture.
Technical Execution — 25%	React/Vite frontend, Node simulation backend, reusable services/hooks/components, explicit simulation state, manual and automatic ticket paths.
UX & Feasibility — 20%	Operator-oriented navigation, live simulation controls, facility context, fixture visualization, analytics, maintenance workflow and local Windows run instructions.
Business & Sustainability — 10%	Water-loss detection, virtual footprint, reduction actions and measurable operational consequences of unresolved leaks.

9. Evidence & Reference Register

Reference	Use in documentation
KOHLER Track 2 case statement supplied for the challenge	Defines objective, key inputs, expected outcome and evaluation criteria.
Final project README	Defines architecture, simulation behavior, analytics, Facility Twin context and submission alignment.
src/data/facility.js	Facility/fixture data model and contextual data.
src/services/api.js	Frontend/backend communication boundary.
src/hooks/useFacilityStream.js	Live stream consumption and reusable telemetry state.
src/pages/Simulation.jsx	Master simulation controls and scenario UX.
src/pages/Analytics.jsx	Water, leak, utilization, workload and footprint analysis.
src/pages/Maintenance.jsx	Manual and automated ticket workflow.
src/pages/FacilityTwin.jsx	Fixture/facility visualization.
src/pages/Sustainability.jsx	Sustainability estimates and reduction actions.
server/index.mjs	Telemetry/ticket simulation behavior and state transitions.

10. Reproducibility & Demo Protocol

The final prototype is intended to be locally runnable and demonstrable without external production infrastructure.

Windows quick start

```
pnpm install
pnpm dev

Open:
http://127.0.0.1:5173/

Optional split-terminal mode:
pnpm dev:backend
pnpm dev:frontend

Backend health:
http://127.0.0.1:8787/api/health
```

Recommended 1–3 minute demonstration sequence

Time	Action	What to demonstrate
0:00–0:20	Open Overview	Facility state, water flow and operational KPIs
0:20–0:40	Open Simulation	Start simulation and show live telemetry
0:40–1:05	Select High Traffic	Usage/attention behavior
1:05–1:35	Select Leak Injection	Critical state → incident → automatic ticket
1:35–1:55	Open Maintenance	Show ticket and manually change/resolve it
1:55–2:20	Open Analytics	Leak trend, projected water loss, workload and footprint
2:20–2:40	Open Sustainability / Facility Twin	Translate operational event into resource impact/location
2:40–3:00	Stop simulation	Show that telemetry/ticket generation freezes

Submission note

The challenge requests a single GitHub repository containing the working model, prompts PDF, 1–3 minute demonstration and maximum four-slide presentation. This document is intended to satisfy the prompts-documentation component and provide reviewers with a transparent development trail.

11. Final Prompt Audit Checklist

Check	Status
Case-study requirements represented	Documented
Telemetry inputs represented	Documented
Continuous leak logic represented	Documented
Attention/critical severity represented	Documented
Automatic maintenance dispatch represented	Documented
Manual ticket creation/removal represented	Documented
Resolved ticket state represented	Documented
Simulation start/stop behavior represented	Documented
Analytics includes leak analysis	Documented
Virtual water footprint represented	Documented
Reduction actions represented	Documented
Facility Twin represented	Documented
AI Copilot grounding/guardrails represented	Documented
Reusable modular architecture represented	Documented
Synthetic-data limitation disclosed	Documented
Prompt-to-code traceability included	Documented

12. Closing Statement

The final system is intentionally positioned as a **prototype of an operational intelligence layer**, not as a claim of production IoT deployment. Its central innovation is the connection between high-frequency facility telemetry, persistent anomaly reasoning, maintenance dispatch, human resolution and sustainability accounting in one reusable workflow.

Prepared for the KOHLER-MITWPU AI Research Lab — Track 2 submission.